

Potential exam questions (AV)

1) What are 2-Dimensional motion models? Explain how they can be

used for image stitching (10)

2) Explain the basic steps involved in choosing compositing surfaces (10)

3) With a neat diagram, explain the projective frustum.

a) 3D to cylindrical coordinates (5)

b) 3D to spherical coordinates (5).

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Be

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~~any thing~~
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4) Write short notes on the following 2D geometric primitives with neat figure. (6)

a) 3D points (2)

b) 2D lines (2)

1) Explain the basic set of 2D plane transformation with neat figure. (6)

5) Explain the general image processing operator in detail (6).

1) Define the following:

a) Compositing.

b) Motion.

6) What is bidirectional reflectance distribution function (BRDF)? (4)

1) With neat figure, discuss the components into which a typical BRDF is split into (6)

7) What is Fourier transforms? Discuss about the closed form equations of Fourier transforms in continuous & discrete domains (6)

1) Differentiate between discrete & fast Fourier transforms (4).

8) What is the need for projective reconstruction? (5)

1) Briefly discuss why does the essential matrix (E) change into the fundamental matrix (F)? (5)

9) Explain hierarchical motion estimation in detail. (10)

10) Explain the various bundle adjustment techniques involved in accurately recovering structure and motion. (10)

11) Explain Fourier based alignment in detail with necessary examples. (10)
12) Explain snakes in detail, what is the main drawback in using snakes? (6)

Q What is the basic solution to overcome the drawback while using snakes? (4)

13) Explain normalized cuts in details with an example. (10)

14) Explain the following systems with necessary examples:
a) diffusion (5)

b) level sets (5)

15) Categorize the various techniques developed for solving pose estimation problem. (10).

16) With a neat diagram, illustrate the ~~pose~~ aperture problem for different image patches. (10)

17) Give the equation for the following and briefly summarize each in few words:
a) TPR and FPR. (4)

b) PPV and Accuracy (ACC). (6)

18) Explain the various techniques in ~~edge~~ detection with necessary examples (10).

19) What is Hough transform? with a neat figure, explain how a 2D line equation is expressed in terms of the normal a & distance ~~to~~ to the origin d . (10)

20) (Two ideas)

Note:

TPR - true positive rate (8) FPR - false positive rate.

PPV - positive predictive value.

Reg. No.	R	A	2	0	1	1	0	2	6	0	2	0	0	6	8
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B.Tech. DEGREE EXAMINATION, DECEMBER 2022
Fifth Semester

18CSE390T – COMPUTER VISION
(For the candidates admitted from the academic year 2020-2021 and 2021 -2022)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part - B** should be answered in answer booklet.

Time: 2½ Hours

Max. Marks: 75

PART – A (25 × 1 = 25 Marks)

Answer ALL Questions

1. Select the sequence of steps involved in computer vision.

- (A) Image acquisition, preprocessing, pattern recognition and decision making
(B) Image preprocessing, acquisition, pattern recognition and decision making
(C) Image acquisition, pattern recognition, preprocessing and decision making
(D) Preprocessing, pattern recognition and decision making

2. What is the degree of freedom of an affine transformation in a 2D image?

- (A) 2
(B) 4
(C) 6
(D) 8

3. A kernel is said to be separable if,

- (A) $r(x, y, u, v) = r_1(x, u)r_1(y, v)$
(B) $r(x, y, u, v) = r_1(x, u)r_2(y, v)$
(C) $r(x, y, u, v) = r_1(x, y)r_2(u, v)$
(D) $r(x, y, u, v) = r_1(x, y)r_1(u, v)$

4. Identify the filter used to invert the blue of an image and remove the extra noise

- (A) Gaussian
(B) Sobel
(C) Wiener
(D) Canny

5. Indirect illumination is taken into account by the following technique

- (A) Diffuse reflection
(B) Specular reflection
(C) Direct illumination
(D) Global illumination

6. Which filter is used to find the brightest point in the image?

- (A) Median filter
(B) Max filter
(C) Min filter
(D) Mean filter

7. Which of the following measures are not used to describe a region in an image?

- (A) Mean and median of gray values
(B) Minimum and maximum of gray values
(C) Number of pixels alone
(D) Number of pixels above and below mean

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1 1 1 1 1

1 2 1 2

1 2 1 2

1 2 1 3

1 1 1 1

1 2 2 2

1 3 2 2

8. In which step of processing, assigning a label to an object based on its descriptor is done? 1 2 2 8
 (A) Object recognition (B) Morphological processing
 (C) Segmentation (D) Representation
9. Gradient magnitude images are more useful in 1 2 2 2
 (A) Point detection (B) Line detection
 (C) Area detection (D) Edge detection
10. Identify the set of connected pixels that lie on the boundary between two regions. 1 3 2 3
 (A) Point (B) Edge
 (C) Color (D) Line
11. Sobel gradient is not good for the detection of 1 1 3 1
 (A) Horizontal lines (B) Vertical lines
 (C) Diagonal lines (D) Edge
12. State the form of control model that adjusts the smooth curve established in the Euclidean plan by moving the curve's point perpendicular. 1 2 3 2
 (A) Snake model (B) Gradient flow model
 (C) Balloon model (D) Geodesic active control models
13. What is said to be evolved from a series of distributed circles towards the final binary segmentation? 1 2 3 2
 (A) Snakes (B) Active contours
 (C) Hough lines (D) Level sets
14. For identifying and classifying the joints in the human body _____ are captured as a set of co-ordinates for each joints. 1 2 3 2
 (A) Keypoints (B) Pairs
 (C) Skeleton (D) Contour
15. State the method which runs a pose detector followed by estimating body joints within the detected bounding boxes. 1 1 3 2
 (A) Top down method (B) Bottom up method
 (C) Level cut method (D) Graph cut method
16. From the given below, select a popular 2D human pose estimation method. 1 1 4 1
 (A) Arpanet (B) Alphanet
 (C) Openpose (D) Close pose
17. Which of the following determines a points 3D position from a set of corresponding image locations and known camera positions? 1 3 4 3
 (A) Level sets (B) Pose estimation
 (C) Triangulation (D) Graph cuts
18. _____ is the process of estimating the 3D structure of a scene from a set of 2D images. 1 1 4 2

- (A) Structure from movement (B) Structure from motion
 (C) Point correspondences (D) Joint correspondences

19. What are needed to estimate motion between two or more images? 1 2 4 4
 (A) Error metric (B) Search techniques
 (C) Error metric and sorting techniques (D) Error metric and search techniques
20. In how many operations, does the fast Fourier transform algorithm compute the transform of a $N \times M$ image? 1 3 4 4
 (A) $O(NM)$ (B) $O(N^2M^2)$
 (C) $O(NM \log N)$ (D) $O(NM \log N)$
21. What technique is used to produce today digital maps and satellite photos? 1 1 5 1
 (A) Image alignment (B) Image mosaicking
 (C) Image stitching (D) Image reconstruction
22. Select the example applicable for high level process in computer vision. 1 1 5 1
 (A) Image sharpening (B) Object recognition
 (C) Noise removal (D) Autonomous navigation
23. _____ helps in providing an image based automatic inspection. 1 1 5 3
 (A) Facial recognition (B) 3D reconstruction
 (C) Machine inspection (D) Automotive safety
24. While performing median filtering, on a 3×3 neighborhood having value (3, 7, 2, 1, 0, 0, 9, 5, 8), what is the median value to be given to the pixel under filter? 1 3 5 4
 (A) 0 (B) 2
 (C) 3 (D) 5
25. The process of simultaneously adjusting pose parameters and 3D point locations for a large collection of overlapping image is called _____. 1 1 5 1
 (A) Photo grammetry (B) Bundle adjustment
 (C) Gap closing (D) Mis-registration

PART – B (5 × 10 = 50 Marks)

Answer ALL Questions

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26. a. Show how 2D transformations are occurring, while an image is formed with suitable examples. 10 2 1 1
- (OR)
- b. Describe the various types of linear filtering techniques available to enhance an image with suitable example. 10 1 1 12
27. a. For an image matching application, illustrate the working principle of SIFT technique. 10 3 2 3

(OR)

- b. For finding lines in an image, illustrate the working of Hough transform algorithm. 10 3 2 3
28. a.i. Why active contour is needed? 2 1 3 4
- ii. How does active contours work? 6 2 3 4
- iii. List the various active contour models available in practice. 2 1 3 1
- (OR)
- b. In the context of human fall detection, explain the working principle of pose estimation and detection method. 10 3 3 3
29. a. With appropriate equations, explain the process of bundle adjustments. 10 2 4 2
- (OR)
- b.i. State the applications of motion model. 4 1 4 1
- ii. How does hierarchical mention estimation is applied to accelerate the search process? 6 2 4 1
30. a. With suitable examples, explain the concept of rotational panoramas. 10 2 5 3
- (OR)
- b.i. People prefer that the find stitched image is upright rather than twisted or tilted. Justify. 6 3 5 4
- ii. List the various motion models in computer vision. 4 1 5 2

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